Decentralized Certification on Bitcoin Using Taproot

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Abstract

Document verification is a crucial part of detecting manipulation and fraud. This process is traditionally centralized, making it costly, fragile, and a single point of failure for an issuing organization. If the issuing organization is compromised, verifiability is lost. In 2009, Bitcoin was deployed as a decentralized network with high security and availability.

Data anchoring methods on Bitcoin, such as `OP_RETURN` or OpenTimeStamp, have limitations in scalability, privacy, and usability. The Bitcoin protocol's Taproot upgrade in 2021 overcomes limitations on the base layer.

This work proposes a new document verification protocol leveraging Taproot's cryptographic tools. It enables the verification of integrity, authenticity of issuance, validity revocation, and signature rotation. Using a Taproot address, it improves the scalability and privacy for a batch of N documents. The protocol is based on an "off-chain verification" model. A rapid Proof of Concept (PoC) prototype will be developed in Python on Bitcoin's testnet/signet network. The objective is to evaluate the efficiency in block size, the on-chain network usage cost, and the off-chain verification efficiency.

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1 Introduction

Certificates issued by an organization are proof of the occurrence of a fact. The verification that this document has not been forged, manipulated, or revoked depends on the issuer. Confidence in the organization rests on this procedure being efficient and accessible.

The verification of a certificate has historically been done in the physical world in a centralized manner. This entails significant costs to maintain high availability and efficiency. With the arrival of the Internet and the proliferation of computers, it was inevitable to bring both issuance and verification into a centralized digital space.

Bitcoin began operating in 2009 as the first permissionless decentralized network. It has not had any censorship events or downtime as of the writing of this document. It allows the execution of limited smart contracts following the UTXO model, as well as data anchoring. Limited by design to reduce the attack surface for denial-of-service and due to the high cost of anchoring bytes of information, it was not considered a scalable solution for document verification.

Various studies and proposals have turned to using other blockchain networks, such as Ethereum. By allowing the execution of Turing-complete smart contracts, they facilitate the validation and verification of certificates. However, they introduce different trust assumptions and greater complexity in consensus, which increases the risk of censorship or rollbacks, compromising verification autonomy.

Bitcoin has a protocol that has evolved over its 16 years of existence. It had two major upgrades accepted by the community, Segwit (2017) and Taproot (2021). These improvements in security, privacy, and blockspace efficiency allow for the expansion of use cases. Utilizing the cryptographic tools enabled by Taproot, we propose a methodology for document verification directly on the base layer of Bitcoin.

2 Literature Review

A Bitcoin transaction is a variable-sized string of Bytes with a theoretical maximum of 1 MB. It has information sections such as the timestamp, version, outputs, etc. We highlight one section in particular: the `scriptPubKey` in the output. The type of transaction is configured depending on the type of `scriptPubKey`.

The standard ones being Pay-to-PublicKeyHash (P2PKH) and Pay-to-ScriptHash

(P2SH), known as legacy transactions. These transactions suffer from a vulnerability called malleability, which was resolved with the arrival of SegWit in 2017. Malleability is a vulnerability that allows the DER encoding of an ECDSA digital signature to be modified without invalidating it, thereby changing the transaction index (txid).

This vulnerability was the primary motivation for developing SegWit in 2017. It modifies the transaction structure by moving the signature information (the `scriptSig`) to a separate section called the Witness. This introduced new transaction types: Pay-to-Witness-PublicKeyHash (P2WPKH) and Pay-to-Witness-Script-Hash (P2WSH).

2.0.1 Data Anchoring Techniques

The use of Bitcoin was initially limited only to the peer-to-peer transmission of monetary value. By maintaining a distributed and consistent ledger across the entire network, the possibility of using it for other applications soon became clear. In the work 'Trusting records: is Blockchain technology the answer?' Lemieux, 2016, it is studied whether this technology is appropriate for preserving the integrity of digital information using various ISO standards. The analysis concludes that, while Blockchain is useful for addressing information integrity in the short term, it does not guarantee the initial reliability of the records and presents several limitations as a long-term solution for authentic digital preservation. The main argument is the vulnerability of digital signature algorithms and hashes like sha256 and ripemd160.

With the exception of Coinbase transactions (where new Bitcoin tokens are issued in each block), Bitcoin transactions do not contain a default space for inserting arbitrary information. The paper 'Decentralized Trusted Timestamping Blockchain' Gao and Nobuhara, 2017 proposes a method to introduce metadata within P2PKH transactions. It encodes information inside a "fake" address, losing control by not having the corresponding private key (sk). By being able to introduce N outputs in a transaction, the proposed method has $20 \times N$ available bytes. This method permanently burns the Bitcoin in each output, making it one of the most expensive.

In the work 'Data Insertion in Bitcoin's Blockchain' Sward et al., 2018, an exhaustive search is made of the ways used to insert data into transactions. It identifies two categories: transactions that use ScriptHash and those that use "fake addresses" as in the case described in the previous paper. It finds that the use of ScriptHashes is a more

complex process to perform. It has advantages in not being limited to 20 bytes; however, its construction is non-standard and needs to be developed at each stage. It concludes that "no universally optimal method exists" as there is a trade-off between the amount of data and the anchoring costs. It also mentions the need for studies to investigate the impact of SegWit, as it is limited to analyzing legacy transactions.

The paper "A Journey into Bitcoin Metadata" Bartoletti et al., 2019 also analyzes techniques for anchoring metadata in Bitcoin and its applications, such as proof of ownership and financial records. It explores in detail whether the different anchoring methods negatively affect Bitcoin's primary function. Finding that one problem is the bloat of the Unspent Transaction Output (UTXO) set. In this regard, it highlights that methods using `OP_RETURN` are those that generate the least impact, occupying less space in relation to the total size of the blockchain.

`OP_RETURN` methods utilize a transaction output, without spending Bitcoin. It is a `scriptPubKey` that contains the `OP_RETURN` opcode, which terminates the script execution. The work "Dominating OP Returns: The Impact of Omni and Veriblock on Bitcoin" Strehle and Steinmetz, 2020 analyzes 32.4 million transactions between 2018 and 2019, finding that the use of `OP_RETURN` comes mainly from 37 services. Primarily: Omni, which anchors financial information related to the use of USDT, and Veriblock, which uses it for the security of other blockchains using Proof-of-Proof. Showing that `OP_RETURN` was established as the standard and dominant method for embedding non-monetary data in the Bitcoin blockchain. It measures their impact on the Bitcoin network, finding that the analyzed transactions contribute approximately 10 GB to the network despite being limited to 80 bytes per output.

The use of legacy transactions for data anchoring has two problems in its design: lack of privacy and the malleability vulnerability. When analyzing the blockchain, the `OP_RETURN` content is clearly identified, losing the privacy of the anchored information. Malleability is a vulnerability that allows an attacker to modify the encoding of a digital signature without affecting its validity, creating two valid transactions but with different transaction IDs (txid).

These two problems are the main reason for the implementation of BIP141 SegWit. In the paper 'Analysis of segregated witness implementation...' Kedziora et al., 2023, the authors study the impact after the softfork's application. It finds benefits in separating the

digital signature from the transaction's data block into another space called the Witness: efficiency and security. A SegWit transaction is 75% less costly and solves the malleability problem. It finds that data anchoring via the Witness expands the limitation in Bytes. While they do not affect the bloating of the UTXO set, they need to be fully declared and spent to be anchored to the Blockchain. Concluding that the implementation was successful and opens the door to new types of applications with more versatile Smart Contracts.

The development of new cryptographic tools is fundamental to improving performance in Blockchain networks. Studies like Gayoso Martínez et al., 2020, Fang Fang et al., 2020, and Zhang Zhang et al., 2025 analyze the essential technological pillars: hash functions, digital signatures, and Merkle trees. They suggest the use of new digital signature schemes, highlighting Schnorr signatures. Adding privacy and efficiency in validation algorithms, they show their advantage in aggregate, group, and threshold signature schemes. The authors agree that Schnorr is a scheme that meets the non-repudiation requirements in Bitcoin.

These foundations are the basis for a new softfork proposed in BIP341 Taproot; it is an extension of SegWit, considered a new version of it. It proposes a new transaction type, P2TR (Pay-to-TapRoot). Since its activation in November 2021, Taproot has been successfully implemented. Opening the possibility for new applications.

2.0.2 Alternative Systems and Applications (Ethereum)

It has been shown that data can be anchored in Bitcoin in different ways. The limitation in Bitcoin's Script execution also limits the applications built on top of it. To overcome these limitations, new Blockchain networks have been created that change the UTXO design that Bitcoin uses. The one with the most development is Ethereum, which emulates a virtual machine called the EVM.

Different works such as Wang Wang et al., 2019, Li **Li2020DecentralizedPKI**, Pennino Pennino et al., 2021, Palmisano Palmisano et al., 2022, Mishra Mishra et al., 2025, and Harer Härer and Fill, 2022 make use of Smart Contracts to attest information such as certificates, dates, and XML structures, respectively. Sophisticated certification and attestation in Smart Contract-based architectures depend on costly execution (gas fees) and programmable complexity (Solidity, EVM) to manage the logic of certificate verifi-

cation, identity, and permissions. The applications vary, but they all make use of data anchoring.

A case of special interest are the works related to the institutional use of document verification. In 'Blockchain and Higher Education Diplomas' Castro and Au-Yong-Oliveira, 2021, the need to eradicate fraud and falsification of diplomas and the sale of fraudulent certificates, a global market estimated at 2 billion USD, is developed. The study notes that, as a radical innovation, blockchain adoption must overcome technical barriers (scalability, error correction, complexity) and cultural/social barriers (lack of standards, resistance to change).

'A Systematic Literature Review on Blockchain-Based Systems for Academic Certificate Verification' Rustemi et al., 2023 identifies that most existing platforms are implemented on the Ethereum blockchain and services are executed via Smart Contracts. The immutability of Smart Contracts is a challenge, as the revocation of a diploma implies a change in the hash value, which challenges the student's identity and is an inherently difficult or impossible process in the blockchain system. Verification is the only process that can be executed completely without third-party intervention, while generation and revocation remain challenging.

In summary, the literature presents a clear trade-off: legacy Bitcoin solutions (like `OP_RETURN`) lack privacy and scalability, while Smart Contract solutions (like Ethereum) are computationally expensive.

Taproot, with its efficient Schnorr signatures and MAST (Merkleized Abstract Syntax Trees), allows encoding complex verification logic (similar to an SC), but does so with a minimal transaction footprint on the Bitcoin blockchain, avoiding the high operational cost of Ethereum's EVM and overcoming the data capacity limitations of old Bitcoin methods like `OP_RETURN`.

3 Motivation

The motivation for this work is the fragility of centralized verification models. If an issuing organization disappears, the verifiability of its documents is lost. The reliability of a decentralized network lies in its consensus protocol. Bitcoin is based on Proof of Work (PoW), making it a secure, immutable, and denial-of-service (DoS) resistant network.

These characteristics are not as strong or are not present in other networks. Solutions in document verification utilize other networks, such as Ethereum, due to the versatility of their Turing-complete scripts and their different scalability trade-offs compared to Bitcoin.

With Taproot, security and blockspace efficiency are improved. This enables new applications on the main layer (layer 1). Our proposal does not require another protocol upgrade, nor does it resort to a second-layer blockchain or networks other than Bitcoin.

4 Problem Statement

Decentralized solutions for document verification have security problems inherent to the network on which they are deployed. On Bitcoin, they contain privacy issues or are inefficient in cost and blockspace in solutions using protocols such as OpenTimestamps, Blockcerts, `OP_RETURN`, among others.

Taproot applies a new, more efficient, and more private Schnorr digital signature scheme for multisignature security schemes. It also allows committing multiple scripts into a single output using Merkleized Abstract Syntax Trees (MAST), improving costs and blockspace usage. This combination has not been previously utilized to create a protocol for document verification.

4.1 Research Questions

This work aims to answer:

- How can the tools applied by Taproot be leveraged to deploy a decentralized verification protocol?
- To what extent does using MAST improve privacy, scalability, and spatial/computational complexity?
- What is an appropriate methodology to add signature rotation and certificate revocation to the protocol?

4.2 Specific Objectives

To answer the research questions, the following objectives are set:

a) Design software that creates the Merkle structure of scripts where the integrity of the issued documents is verified.

b) Design a software component that allows ownership verification using the digital signature on issued documents.

c) Design a software component that allows for document signature key rotation.

d) Design a software component that tracks revocation to verify the validity of issued documents.

e) Deploy a Proof of Concept (PoC) on a Bitcoin test network.

f) Measure the scalability, time/space complexity, and network costs as a function of the vBytes used in the transaction.

5 Methodology

To meet the objectives, we propose the following methodology divided into three phases: (1) Protocol and architecture design, (2) Proof of Concept prototyping, and (3) Experimental validation and analysis.

5.1 Phase I

In this phase, objectives a, b, c, and d are developed. We highlight the components in our protocol's scheme:

- The issuer corresponds to the organization that will provide the document and an additional file, which we call a 'proof of validation', to a user.
- The verifier, who receives the document and the user's proof of validation. Verifies the document's validity.

Only access to the public Bitcoin blockchain history is necessary, in addition to the document and the proof of validity. The complete verification consists of the following components:

- Integrity verification, ensures the document's content has not been manipulated.

- Ownership verification, ensures it is unequivocally issued by the organization.
- Revocation verification, ensures the document's validity has not been revoked.

If all these verifications are met, the document is considered valid. Taproot allows us to evaluate a number N of certificates, ordered in batches, in a single transaction. The computational procedure for verification occurs off-chain; that is, it is not executed on the Bitcoin network.

5.1.1 Integrity Verification

To preserve the privacy of the document's content, integrity verification follows a procedure using the sha256 cryptographic hash as a base.

The cryptographic architecture for anchoring the hashes of a batch of certificates will be defined following the BIP341 protocol to obtain a valid Merkle root, adhering to the structure shown in Figure 1. Each Merkle Leaf contains the bytes of each document's hash within provably unspendable scripts; that is, they do not enable the spending of the transaction. Using the 'Tweak' procedure with the issuing organization's single Schnorr public key for Bitcoin, a Taproot address is derived.

In this process, the issuer must provide the necessary information in the proof of validation to be able to reconstruct the Taproot address. For verification, the protocol demonstrates that, by calculating the certificate's hash, the same Taproot address of the transaction in which the batch is anchored is derived.

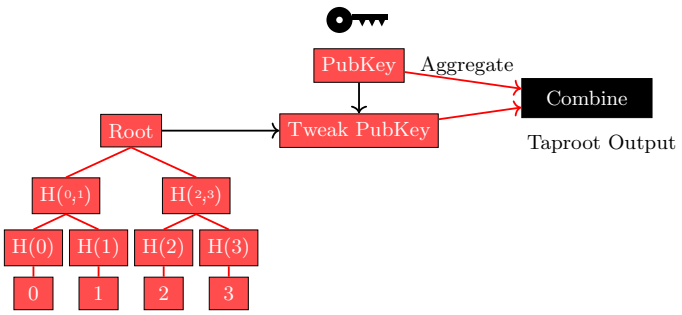

Figure 1: MAST structure for certificate anchoring

By containing provably unspendable scripts, the verifier is limited to comparing the Taproot address. They do not obtain any information about other documents. It does not reveal the content, nor can it spend the Taproot transaction output via the script-path.

5.1.2 Ownership Verification

We propose a scheme that contains two different Schnorr key-pairs. One for signing documents, $Keys_{firma}$, and another for maintaining the Bitcoin funds, $Keys_{btc}$. The reason is to have a separation of concerns and to improve exposure risks.

Because the batches of documents to be signed can be of variable size, the private key Sk_{firma} can be managed on the organization's own server. The digital signature is added to the proof of validity given to the user. The public key Pk_{firma} is exposed in a second output of the Taproot transaction via `OP_RETURN`.

The protocol validates the digital signature declared in the proof of validity against the document and the public key Pk_{firma}.

5.1.3 Signature Rotation

Signature rotation refers to the protocol's ability to change the use of digital signature keys P_{firma} without affecting the validity of previously issued documents. If a new batch changes the signature key, the new public key $Pk_{firma'}$ is anchored in the `OP_RETURN` of the transaction. Maintaining an order of each batch with its corresponding public key Pk_{firma}.

5.1.4 Document Revocation

The issuing organization may be interested in revoking the validity of a set of issued documents. A revocation scheme is proposed by declaring a validity state in a child transaction that spends the Taproot output of the issued document batch. The organization must use the Key-Path, using its private key Sk_{btc}, to spend the Taproot transaction output. The new child transaction must have an `OP_RETURN` output that contains the revocation status.

The verifier queries the last chained transaction that spends the parent transaction's output. It looks for the corresponding state in the `OP_RETURN` output according to its order in the Merkle tree. A single bit in its position indicates the document's revocation.

The revocation status update is for the entire batch of documents in the parent transaction. If an update is needed, it generates an extra network cost that must be considered in the protocol.

5.2 Phase II

A second phase of the protocol is to deploy a Proof of Concept (PoC) on the Bitcoin network. To avoid incurring bitcoin token expenses for testing, it is proposed to set up nodes with Signet/Testnet networks.

For rapid prototyping, scripts will be developed in the Python language. The Bitcoin node will be configured on a Linux Debian operating system under Bitcoin-Core version 30.0. Documenting the entire process so that any verifier can run tests without needing third-party services.

The formats in which an issuer can deliver a proof document in '.pdf' format and the proof of validity as a '.json' file to a user will be evaluated.

5.3 Phase III

With a functional PoC, experimental tests will be conducted to obtain performance metrics by varying the size of the document batches N and revocations. Two metrics will be obtained to measure the protocol's performance:

- Scalability: dependent on the batch size N, the time/space complexity of the proof of validity and the validation will be measured. Theoretically bounded at $O(\log N)$ by using Merkle trees.
- Fee for Bitcoin network usage: measured in vBytes as a function of the batch size N and revocations.

With the results obtained, we seek to pose a trade-off between privacy, scalability, and network costs when using Taproot in a decentralized document verification protocol.

6 Timeline

The following timeline details the stages described in 5 and is structured into four phases:

- Weeks 1-4: Construction of the decentralized certification protocol using Taproot on Bitcoin.
- Weeks 5-6: Development of a Proof of Concept (PoC) and its deployment on the Bitcoin test network.

- Weeks 7-8: Evaluation of the performance and security of the implemented protocol.
- Weeks 9-10: Documentation of findings and publication of results.

6.0.1 Gantt Chart Representation

Figure 2 illustrates the proposed timeline using a Gantt chart, providing a clear visual representation of the phases and their durations.

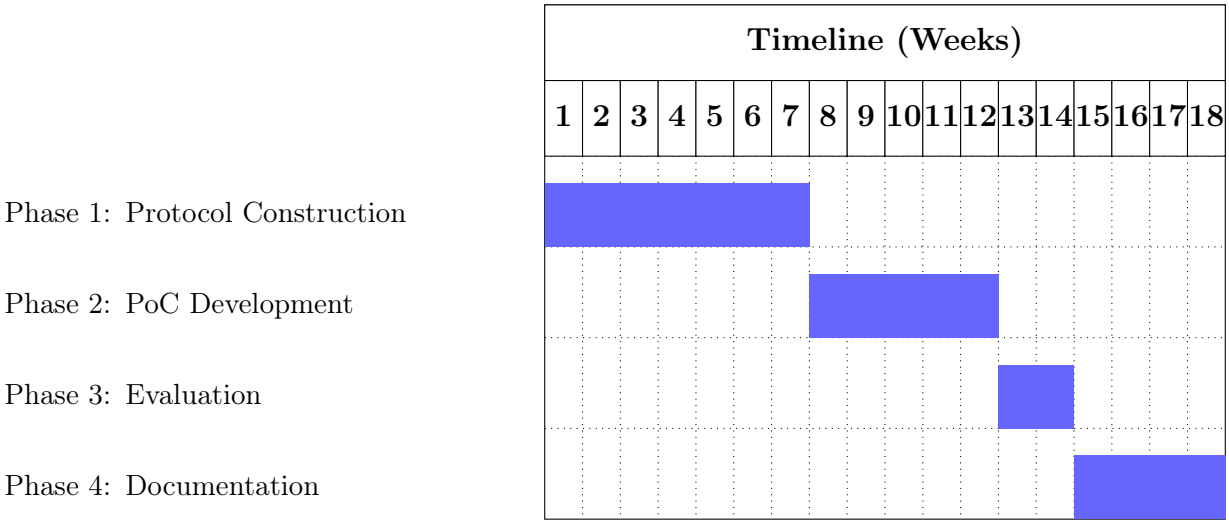

Figure 2: Gantt Chart of the Proposed Timeline

7 Conclusion

A decentralized certification protocol has advantages compared to traditional centralized systems. Availability, denial-of-service resistance, and immutability security are characteristics of the Bitcoin network. Methods studied in the literature show that Bitcoin has limitations regarding scalability, privacy, and applications in document verification. Solutions like the use of `OP_RETURN` and OpenTimeStamp protocols have their limitations. With the Taproot upgrade operational on the network since 2021, Bitcoin overcomes these limitations. It was identified that Taproot has not been studied in the literature for the use case of document verification.

The protocol proposed in this work leverages the cryptographic tools used in Taproot. It allows integrity verification via the hash anchored in the Merkle tree and document ownership via Schnorr digital signatures. It enables the revocation of document valid-

ity and proposes key rotation by chaining transactions that spend the previous output. Preserving privacy by not revealing information from other documents in the batch.

The proposed protocol does not consider the validity of the content of the issued documents. The revocation of validity remains the responsibility of the issuing organization. It does not consider the preservation of the document and its proof of validity once issued and delivered to the end-user. The study of long-term preservation and proof-of-validity mechanisms remains for future work. As well as the implementation of penalties and incentives for users who present invalid documents as valid.

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